

1. In class we had written down an expression for $\Pi(k^2) - \Pi(0)$:

$$\Pi(k^2) - \Pi(0) = \frac{2\alpha}{\pi} \int_0^1 dx \, x(1-x) \ln \frac{m^2 - x(1-x)s}{m^2},$$

where, $s = k^2$. Perform the x integration and find an expression for this quantity in both the regions $0 < s < 4m^2$ and $s > 4m^2$ by analytic continuation wherever needed. What is the imaginary part of this diagram. What physical process is this imaginary part indicative of? (Please look at the explanatory note added to the canvas site where I have discussed how to evaluate such integrals. for different regions of the external parameters)

2. Consider the following theory which describes the interaction of 2 scalar fields, ϕ and χ :

$$L = \frac{1}{2}\partial_\mu\phi^2 - \frac{1}{2}m^2\phi^2 + \frac{1}{2}\partial_\mu\chi^2 - \frac{1}{2}\mu^2\chi^2 - g\phi^2\chi$$

- (a) Find the self energy, $-i\Pi(p^2)$, of the χ particle to one loop. Evaluate both the real and imaginary parts.
- (b) Find the decay rate, Γ , at tree level for the χ particle to decay to two ϕ particles. Show explicitly the relation between Γ and the imaginary part of $\Pi(\mu^2)$.